

A Literature-Implied Answer to the $\Phi_1 \setminus \Phi_2$ Negative-Squares Conjecture

Run `fa_banach_001`: candidate status note

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Status. *Literature-implied answer (full conjecture; candidate likely valid).* The supporting theorem predates the conjecture and does not mention it. The short implication below appears not to have been identified in the bounded search described at the end of this note.

1 Source question

Golinskii, Malamud, and Oridoroga define, for a continuous real function f on $[0, \infty)$,

$$\kappa_n^-(f) = \sup_X n_-([f(|x_i - x_j|)]_{i,j}),$$

where X ranges over finite subsets of $\mathbb{R}^n$ and n_- counts negative eigenvalues with multiplicity. On PDF page 3 of [1] they state:

For each function $f \in \Phi_1 \setminus \Phi_2$, the relation $\kappa_2^-(f) = +\infty$ holds.

Here $f \in \Phi_n$ means that $x \mapsto f(|x|)$ is positive definite on $\mathbb{R}^n$.

2 Supporting classification

We use Theorem 3.2 of Sasvari [2], on journal pages 229–231 (PDF pages 7–9 of the local primary scan). In the special form needed here it says:

Theorem 1 (Sasvari’s bounded classification). *Let G be a locally compact abelian group. If a bounded continuous Hermitian function h on G has exactly $k < \infty$ negative squares, then*

$$h(x) = \int_{\widehat{G}} \gamma(x) d\mu_+(\gamma) - \int_{\widehat{G}} \gamma(x) d\mu_-(\gamma),$$

where μ_+ and μ_- are finite positive measures, μ_- is concentrated at exactly k distinct characters, and μ_+ gives those characters no mass. Conversely, such a representation has exactly k negative squares.

The introduction of Sasvari’s later paper [3, p. 319] independently restates this result: bounded functions with k negative squares are precisely Fourier transforms of measures that are negative at k points of the character group and nonnegative outside those points.

3 The implication

Lemma 1. *Let σ be a finite positive rotation-invariant Borel measure on $\mathbb{R}^2$ with $\sigma(\{0\}) = 0$. Then every line L through the origin has $\sigma(L) = 0$.*

Proof. If $\sigma(L) > 0$, then $\sigma(L \setminus \{0\}) > 0$. Choose countably many rotations taking L to pairwise distinct unoriented lines L_j . The sets $L_j \setminus \{0\}$ are pairwise disjoint and, by rotation invariance, all have the same positive measure. Their union would have infinite measure, contrary to finiteness of σ . $\square$

Theorem 2. *For every $f \in \Phi_1 \setminus \Phi_2$ one has $\kappa_2^-(f) = +\infty$.*

Proof. Set

$$g(x) = f(|x|), \quad x \in \mathbb{R}^2.$$

Because $f \in \Phi_1$, the 2×2 positive-definiteness inequality on the real line implies $|f(t)| \leq f(0)$; hence g is bounded. It is also continuous and Hermitian.

Suppose, for a contradiction, that $\kappa_2^-(f) = k < \infty$. Since $f \notin \Phi_2$, we have $k \geq 1$. The supremum defining k is over integers, so some finite Schoenberg matrix has exactly k negative eigenvalues; thus g has exactly k negative squares in Sasvari's sense. Identify $\widehat{\mathbb{R}^2}$ with $\mathbb{R}^2$. Theorem 1 gives a finite signed measure

$$\mu = \mu_+ - \mu_-, \quad g(x) = \int_{\mathbb{R}^2} e^{i\langle x, \xi \rangle} d\mu(\xi), \quad (1)$$

whose negative part μ_- is supported at exactly k points.

For each planar rotation R , radially gives $g(Rx) = g(x)$. Uniqueness of Fourier–Stieltjes transforms of finite measures in (1) implies $R_*\mu = \mu$ (replacing R by its transpose, which runs through the same rotation group). Uniqueness of the Jordan decomposition then gives $R_*\mu_+ = \mu_+$ and $R_*\mu_- = \mu_-$. The support of μ_- is therefore a finite rotation-invariant subset of $\mathbb{R}^2$. Every nonzero point has an infinite rotation orbit, so

$$\mu_- = a\delta_0 \quad \text{for some } a > 0. \quad (2)$$

Moreover $\mu_+(\{0\}) = 0$, because μ_+ and μ_- are the mutually singular Jordan parts.

Let $\pi(\xi_1, \xi_2) = \xi_1$. Restricting (1) to the first coordinate axis shows that

$$f(|t|) = g(t, 0) = \int_{\mathbb{R}} e^{its} d(\pi_*\mu)(s).$$

The left side is positive definite on $\mathbb{R}$, since $f \in \Phi_1$. Bochner's theorem and uniqueness of Fourier–Stieltjes transforms therefore force the signed measure $\pi_*\mu$ to be positive.

On the other hand, $\pi^{-1}(\{0\}) = L^\perp := \{(0, s) : s \in \mathbb{R}\}$. Apply Lemma 1 to the finite rotation-invariant measure μ_+ : $\mu_+(L^\perp) = 0$. Using (2),

$$(\pi_*\mu)(\{0\}) = \mu(L^\perp) = \mu_+(L^\perp) - a = -a < 0,$$

contradicting positivity of $\pi_*\mu$. Hence no finite k is possible, and $\kappa_2^-(f) = +\infty$. $\square$

4 Identification and scope

The proof uses no regularity, decay, or support hypothesis beyond $f \in \Phi_1 \setminus \Phi_2$. It therefore establishes the full conjecture, not merely the additional-measure-hypothesis cases proved in the source paper. The mathematical input is nevertheless Sasvari's 1985 theorem; the new content of this packet is the direct radial-symmetry and line-projection identification. For that reason its provenance is *literature-implied answer*, not an original full result.

A bounded novelty/status search was run on 13 August 2026 in the run's registry, solution, attempt, and proof-gap indexes and on the web using the exact source title, the notation $\kappa_2^-(f)$, variants of " Φ_1 , Φ_2 , negative squares", and Sasvari's title and DOI. No later paper explicitly claiming this conjecture as resolved was found. That negative search is bounded and is not a claim of exhaustive bibliographic novelty.

References

- [1] L. Golinskii, M. Malamud, and L. Oridoroga, *Radial positive definite functions and Schoenberg matrices with negative eigenvalues*, arXiv:1502.07179 (2015), conjecture on PDF p. 3.
- [2] Z. Sasvari, *On bounded functions with a finite number of negative squares*, Monatsh. Math. **99** (1985), 223–234, doi:10.1007/BF01295156; Theorem 3.2, pp. 229–231.
- [3] Z. Sasvari, *Characterization of locally bounded functions with a finite number of negative squares*, Acta Sci. Math. (Szeged) **53** (1989), 319–327; classification restated on p. 319.